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BENGALURU, INDIA

WHERE THE

WORLD COMES

TO LEARN

TRADITIONAL MEDICINE AND HEALTH MANAGEMENT- BBT755C

Dr.Ashalatha

Asst Prof G I

Department of Biotechnology



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Overveiw

Acharya stands as a beacon of excellence in higher education, boasting a legacy of academic distinction since its establishment in 1990. We offer a transformative educational experience, fostering holistic development, nurturing innovation and providing world-class facilities to ensure an enriching journey for our students.

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MODULE – 3 (8 hours) PLANT CHEMISTRY & PHARMACOLOGY

The chemical and physical structure, properties and functions of the main classes of secondary plant chemicals.

The dynamics and kinetics of medicinal substances upon the human body – remedy absorption, distribution, metabolism, excretion, and sensitivity.

The toxicology of commonly used medicinal plants: side effects, cautions and contraindications.

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1. The chemical and physical structure, properties and functions of the main classes of secondary plant chemicals.

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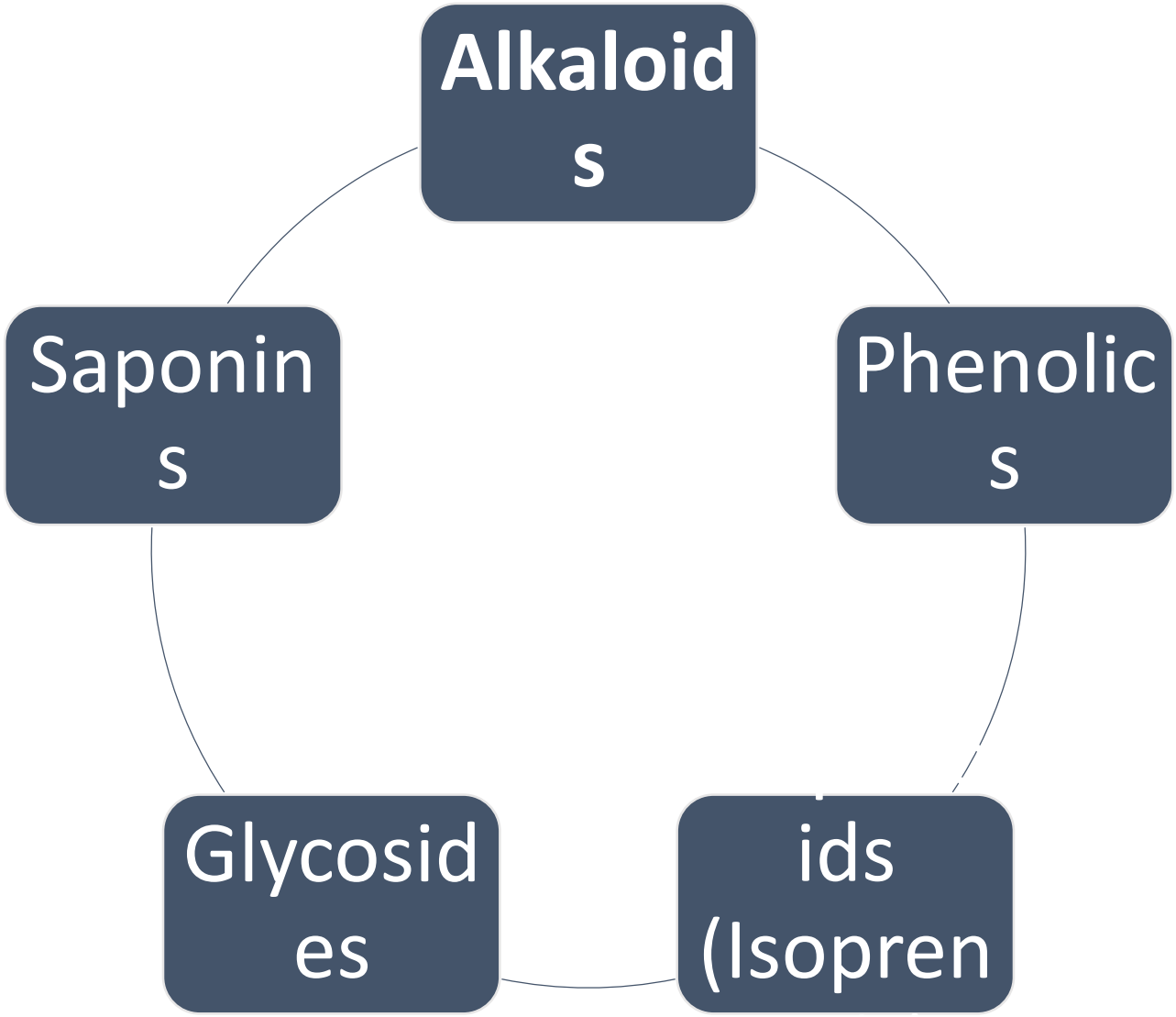
Secondary Plant Chemicals

- Secondary plant chemicals, also called **secondary metabolites**, are organic compounds produced by plants that are **not directly involved in growth, development, or reproduction** (unlike primary metabolites such as carbohydrates, proteins, and lipids).
- Instead, they play crucial roles in **defense, signaling, and ecological Interactions**.
- **Secondary plant chemicals are the "defense and survival toolkit" of plants**, with huge importance in medicine, agriculture, and industry.
- **Characteristics**
  - Not essential for basic survival but important for **adaptation**.
  - Usually produced in specific plant parts (roots, leaves, bark, seeds, flowers).
  - Often occur in **small quantities** compared to primary metabolites.
  - Help plants **survive in harsh environments** by defending against herbivores, pathogens, and competition.

Major Classes of Secondary Plant Chemicals

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Alkaloids

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- Alkaloids are diverse, nitrogen-containing natural compounds, often found in plants, that possess a basic, alkali-like nature. Their chemical structures are variable but typically include a nitrogen atom within a ring system.
- Functionally, alkaloids act as protective substances, metabolic regulators, and reservoirs for nitrogen in plants, while also exhibiting a wide range of pharmacological effects on humans and animals, leading to their extensive use as medicinal agents.

Alkaloids

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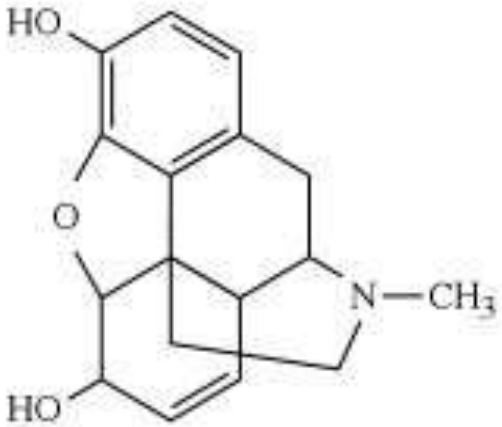
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- Physical Properties
  - Most are crystalline solids; some liquids (e.g., nicotine).
  - Soluble in organic solvents; alkaloid salts are water-soluble.
  - Bitter taste, toxic in high doses.
- Chemical structure:
  - Contain one or more nitrogen atoms, usually in a heterocyclic ring.
  - Derived from amino acids (tryptophan → indole alkaloids, tyrosine → isoquinoline alkaloids).
  - Basic in nature → form salts with acids.
- **Examples & Functions**:
  - **Morphine ($C_{17}H_{19}NO_3$)** → opioid analgesic (painkiller).
  - **Quinine ($C_{20}H_{24}N_2O_2$)** → antimalarial.
  - **Caffeine ($C_8H_{10}N_4O_2$)** → central nervous stimulant.
  - **Nicotine ($C_{10}H_{14}N_2$)** → neurotoxin, natural insecticide.

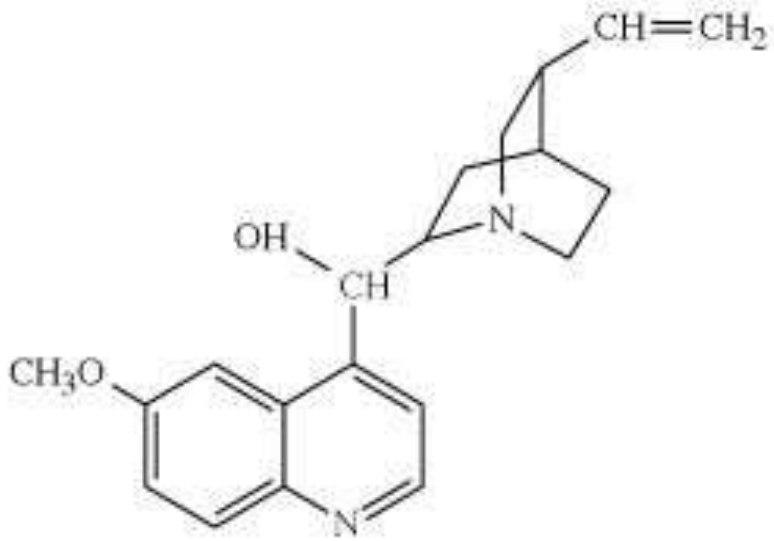
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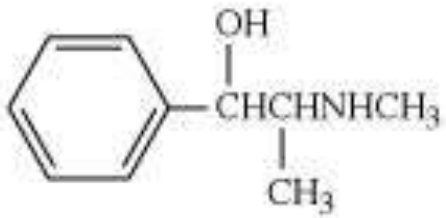
Alkaloids



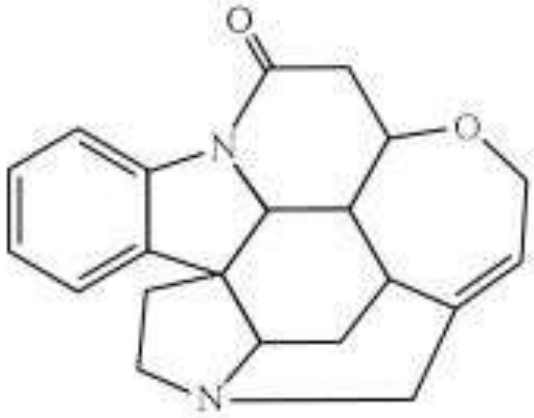
morphine



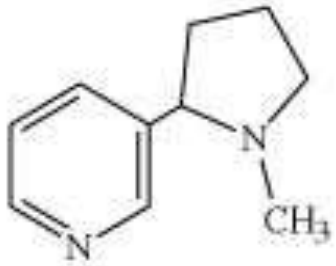
quinine



ephedrine



strychnine



nicotine

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Terpenoids

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- Terpenoids are natural organic compounds with diverse structures and functions, built from 5-carbon isoprene units and containing additional functional groups, often oxygen.
- They contribute to plant scents, colors, and flavors (e.g., camphor, cinnamon), serve as essential nutrients (like provitamin beta carotene), and possess significant bioactivity, making them valuable in medicine and industry.

Properties

- **Aromatic Qualities:**
- Many terpenoids are volatile and possess strong scents, contributing to the characteristic aromas of plants like eucalyptus, cinnamon, and ginger.
- **Color:**
- Terpenoids, such as carotenoids (like provitamin beta carotene), are responsible for vibrant colors in nature, like the yellow of sunflowers and the red of tomatoes.
- **Bioactivity:**
- A key property of terpenoids is their significant biological and pharmacological activity, making them subjects of great interest in medicinal chemistry.

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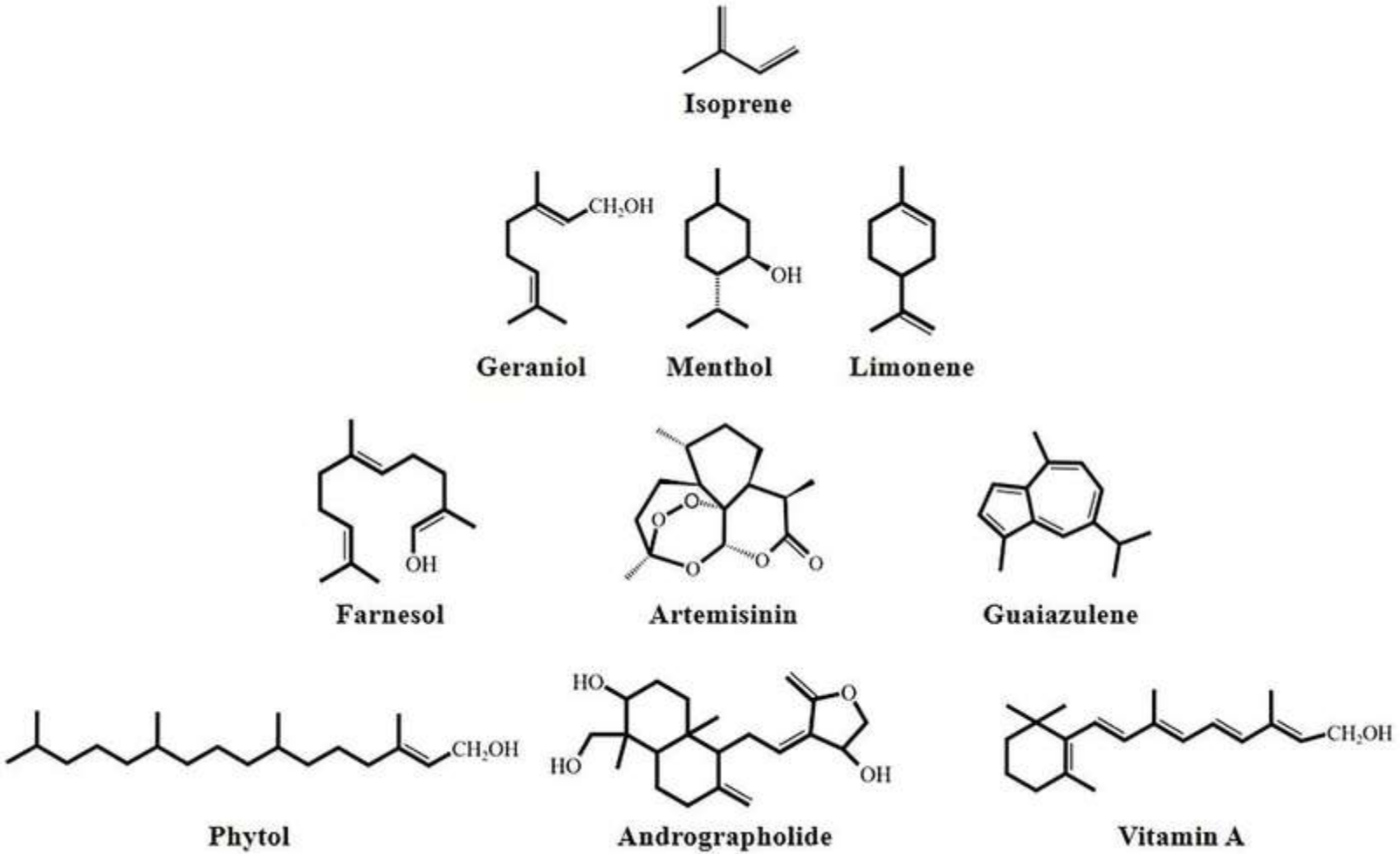
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Properties

- Derived from isoprene unit (C5H8).
  - Built by the mevalonate pathway and MEP/DOXP pathway.
- Classification:
  - Monoterpenes (C10) → essential oils.
  - Sesquiterpenes (C15) → volatile fragrances.
  - Diterpenes (C20) → resins, medicinal.
  - Triterpenes (C30) → steroids, saponins.
  - Polyterpenes → natural rubber.
- Physical properties:
  - Volatile, hydrophobic, strongly aromatic.
  - Many are color-giving pigments (carotenoids).
- **Examples & Functions**:
- **Menthol (C10H20O)** → cooling sensation, insect repellent.
- **β-carotene (C40H56)** → orange pigment, vitamin A precursor.
- **Taxol (C47H51NO14)** → anticancer agent.
- **Rubber (polyisoprene)** → elasticity, commercial use.

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Glycosides

- **Chemistry**:
  - Composed of a **sugar (glycone)** + **non-sugar moiety (aglycone)**.
  - Glycosidic bond links them.
  - Classified as **cyanogenic glycosides, cardiac glycosides, anthraquinone glycosides**.

-Cardiac glycosides are a class of organic compounds and medications derived mostly from plants like Digitalis purpurea that strengthen heart muscle contractions and slow the heart rate=- Found in foxglove, lily-of-the-valley

-Cyanogenic glycosides are a class of chemical compounds found in many plants that release highly toxic hydrogen cyanide (HCN) when broken down by enzymes-Found in Stone Fruits, cassava, bitter almond

-Anthraquinone glycosides are a class of organic compounds found in certain plants that are primarily known for their potent stimulant laxative effects on the human digestive system- found in Senna, aloe vera

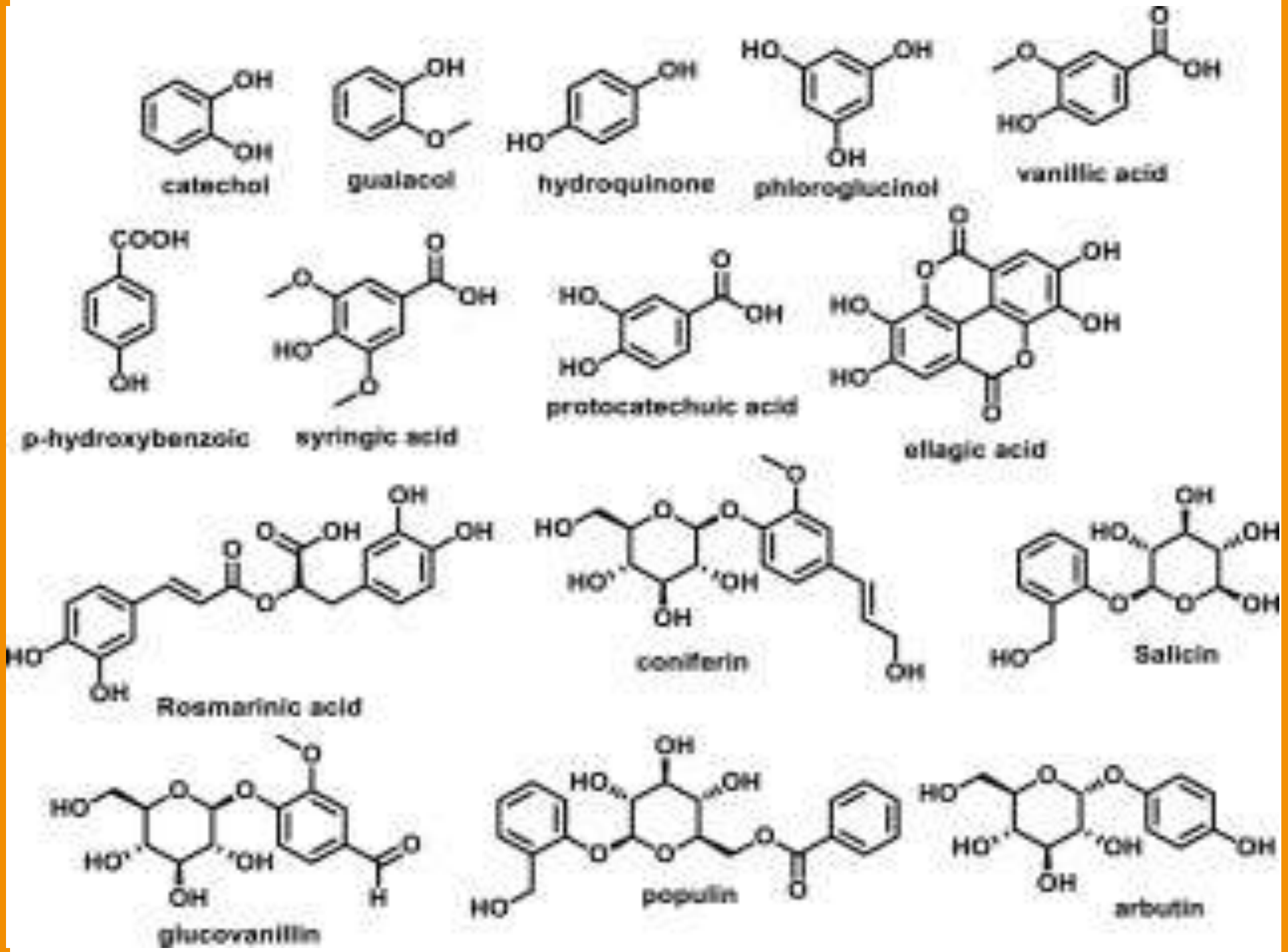
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- **Physical properties**:
  - Water-soluble (due to sugar part).
  - Mostly crystalline, tasteless or bitter.
- **Examples & Functions**:
  - **Amygdalin (C20H27NO11)** → cyanogenic glycoside in bitter almonds, releases HCN for defense.
  - **Digitoxin (C41H64O13)** → cardiac glycoside from foxglove, regulates heartbeat.
  - **Anthraquinone glycosides** → laxative action (senna).

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Saponins

Saponins are a large class of naturally occurring plant glycosides named for their distinct ability to produce a soap-like lather when shaken in water.

Structural Characterization

- Amphiphilic Nature: They contain a fat-soluble non-sugar core (aglycone) attached to water-soluble sugar chains
- Triterpenoid Saponins: Possess a 30-carbon core; highly prevalent in dicotyledonous plants like soybeans and alfalfa.
- Steroidal Saponins: Possess a 27-carbon core; commonly found in monocotyledonous plants like yams and fenugreek.
- Common Sources
- Legumes: High concentrations are found in chickpeas, lentils, peanuts, and soybeans
- .Ginseng Root: Contains specialized triterpenoid saponins called ginsenoids, prized in traditional medicine.
- Quinoa: The bitter outer coating of the grain is rich in saponins, requiring thorough rinsing before cooking
- .Soapbark Tree: Quillaja saponaria bark yields industrial-grade saponins used as commercial foaming agents.

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Saponins

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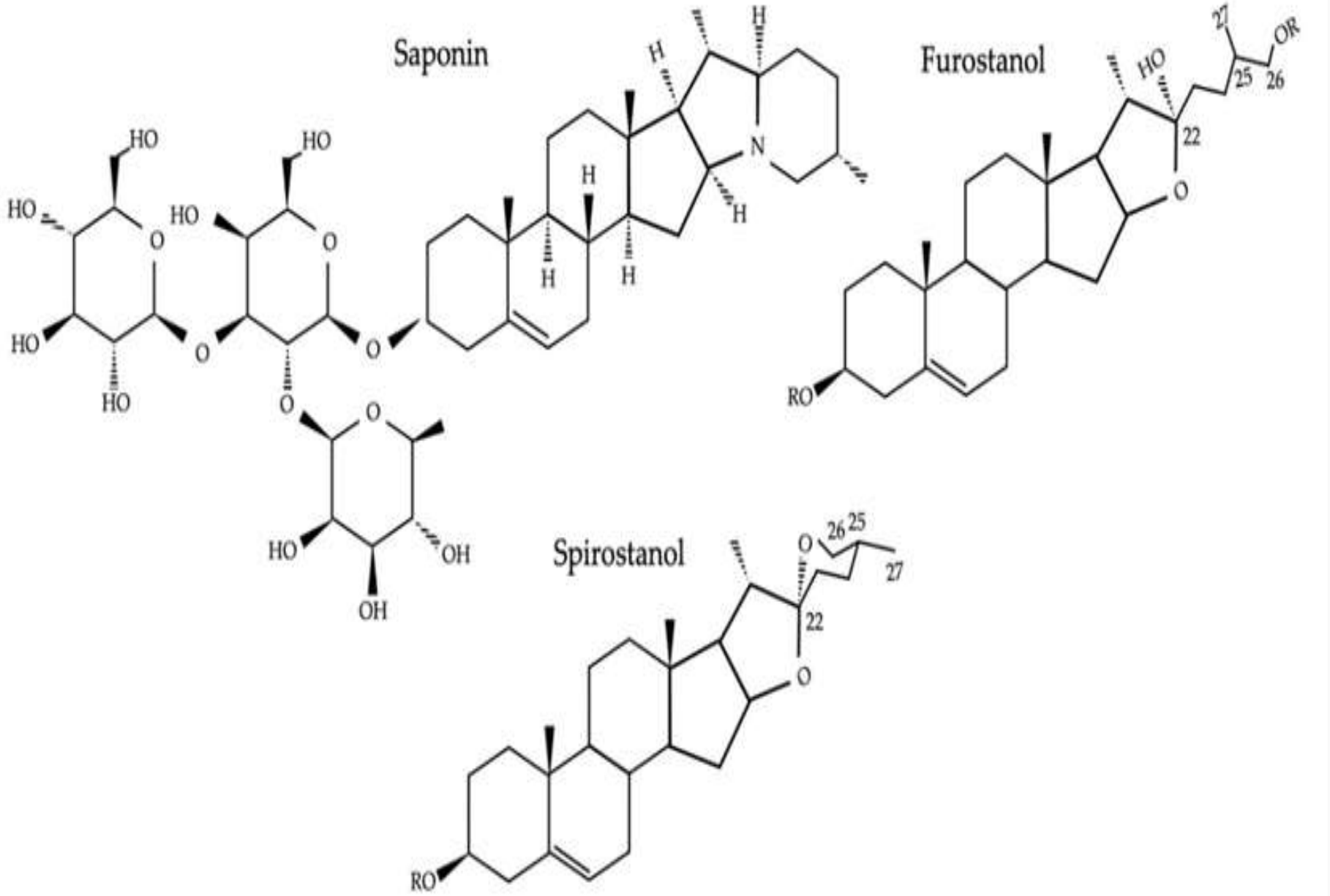
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- **Physical properties**:
  - Bitter, amorphous solids.
  - Amphipathic → dissolve in water and form micelles.
  - Hemolytic activity (break RBC membranes).
- **Examples & Functions**:
  - Found in **soybean, quinoa, ginseng, fenugreek**.
  - Antifungal and antimicrobial.
  - Reduce cholesterol by binding bile acids.
  - Plant defense against insects.

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Saponins



Phenolics

Phenolics are a vast and diverse class of plant secondary metabolites characterized structurally by possessing at least one aromatic benzene ring directly bonded to one or more hydroxyl (-OH) groups

Physical & Chemical Properties:

- Antioxidant Activity: The aromatic ring delocalizes unpaired electrons, allowing phenolics to scavenge free radicals and form stable radical intermediates.
- UV Light Absorption: The highly conjugated double-bond systems absorb ultraviolet light, acting as a natural sunscreen for plant tissues.
- Protein Binding: Tannins and larger polyphenols physically cross-link with proteins, precipitating them out of solution and causing an astringent sensation.
- Acidity: The hydroxyl group attached to the aromatic ring is weakly acidic because the resulting negative charge is stabilized by the resonance of the benzene ring.

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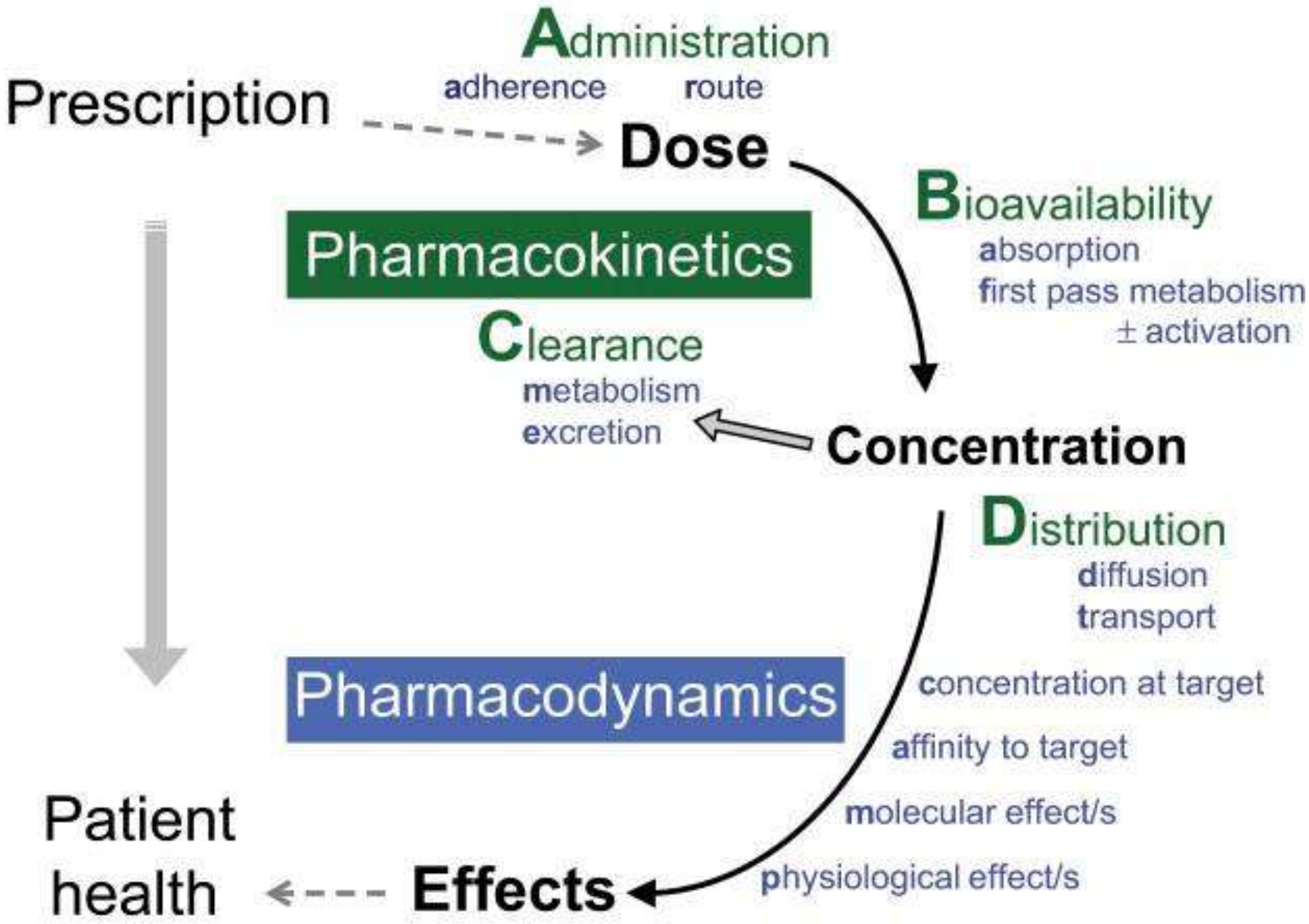
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Biological Roles

- Plant Defense: Protects the host plant against herbivores, fungal pathogens, insects, and oxidative stress.
- Pigmentation: Anthocyanins (a subclass of flavonoids) provide red, purple, and blue coloration to flowers and fruits to attract pollinators.
- Human Health Benefits: Associated with anti-inflammatory, antimicrobial, and cardiovascular protective effects when consumed in dietary fruits, vegetables, green tea, and red wine.

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Absorption

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- **Definition:**
  Absorption is the process by which a drug enters the bloodstream from its site of administration.
- **Key Factors Affecting Absorption:**
  - **Route of administration:** Oral, intravenous, intramuscular, subcutaneous, transdermal, etc.
  - **Physicochemical properties:** Lipid solubility, molecular size, ionization, and formulation.
  - **Gastrointestinal factors:** pH, food, gastric motility, presence of enzymes.
  - **Blood flow** to the absorption site.
- **Mechanisms of Absorption:**
  - **Passive diffusion** – movement from high to low concentration (most common).
  - **Facilitated diffusion** – requires carrier proteins, no energy.
  - **Active transport** – energy-dependent, against concentration gradient.
  - **Endocytosis and exocytosis** – mainly for macromolecules.
- **Example:**
  - **Lipid-soluble drugs** (e.g., diazepam) absorb faster through cell membranes.
  - **Ionized or large molecules** (e.g., insulin) absorb slower.

Factors Influencing Absorption

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Factor	Description	Example
Physicochemical properties	Molecular size, lipid solubility, ionization, formulation	Lipid-soluble drugs like diazepam absorb faster
Route of administration	IV (100% bioavailability), oral, sublingual, rectal, transdermal, etc.	IV > IM > SC > Oral
pH and ionization (Henderson–Hasselbalch principle)	Non-ionized form crosses membranes easily	Weak acids absorb in stomach; weak bases in intestine
Surface area and blood flow	More surface area & perfusion → better absorption	Small intestine > stomach
Gastrointestinal contents	Food, enzymes, motility affect absorption	Food delays absorption of some antibiotics
Formulation and excipients	Particle size, coating, solubility enhancers	Controlled-release vs immediate-release tablets

Mechanisms of Absorption

- **Passive diffusion** (major route) – along concentration gradient.
- **Facilitated diffusion** – carrier-mediated, no energy (e.g., vitamin B12).
- **Active transport** – energy-dependent, saturable (e.g., levodopa).
- **Endocytosis/pinocytosis** – engulfing of macromolecules (e.g., insulin via vesicles).
- **Bioavailability (F):**
- The fraction of the administered drug that reaches systemic circulation unchanged.

$$F = \frac{AUC_{oral}}{AUC_{IV}} \times 100$$

- **IV route:** 100% bioavailability
- **Oral route:** Subject to **first-pass metabolism** in the liver

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Distribution

- **Definition:**
  Distribution refers to the reversible transfer of a drug from the bloodstream to tissues and organs.
- **Determinants of Distribution:**
  - **Blood flow** to tissues (high in heart, liver, kidneys; low in fat and bone).
  - **Plasma protein binding:** Drugs bound to plasma proteins (like albumin) are inactive; only free drugs are active.
  - **Tissue binding:** Drugs may accumulate in tissues (e.g., tetracycline in bones).
  - **Lipid solubility:** Lipophilic drugs cross membranes easily.
- **Volume of Distribution (Vd):**
  Indicates the extent to which a drug spreads in the body.

$$V_d = \frac{\text{Amount of drug in body}}{\text{Plasma concentration}}$$

- **Example:**
- Drugs with high Vd → Widely distributed in tissues (e.g., chloroquine).
- Drugs with low Vd → Confined to plasma (e.g., heparin).

Metabolism (Biotransformation)

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- **Definition:**
  Drug metabolism is the biotransformation of a drug by organs or tissues (primarily the liver, kidney, skin or digestive tract) so that the drug can be excreted. It is the enzymatic conversion of drugs into more water-soluble compounds for easier elimination. To facilitate removal via feces or urine, the drug compound is altered to become more water-soluble.
- **Primary Site:** Liver (via hepatic enzymes, mainly cytochrome P450 system).
- **Phases of Metabolism:**
- **Phase I (Functionalization reactions):**
  - Oxidation, reduction, hydrolysis
  - Introduces or exposes functional groups (–OH, –NH_2, –SH)
  - **Example: Conversion of codeine to morphine.**
- **Phase II (Conjugation reactions):**
  - Combines the drug (or metabolite) with endogenous molecules like glucuronic acid, sulfate, or glycine.
  - Results in highly polar, inactive metabolites.
  - **Example: Glucuronidation of morphine**.
- **Factors Influencing Metabolism:**
  - **Genetic factors** (e.g., slow/fast acetylators)
  - **Age** (newborns and elderly metabolize drugs slower)
  - **Liver disease**
  - **Enzyme induction/inhibition** by other drugs
  - **Diet and environmental factors**

Phases of Metabolism

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Phase	Reaction Type	Purpose	Example
Phase I (Functionalization)	Oxidation, reduction, hydrolysis	Introduces/exposes functional groups; may activate/inactivate drug	Oxidation of codeine to morphine
Phase II (Conjugation)	Glucuronidation, sulfation, acetylation, methylation	Combines drug with endogenous substrate → polar, inactive, easily excreted	Morphine glucuronide, acetaminophen sulfate

PHASE I – Functionalization Reactions

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- Phase I biotransformations alter the chemical structure of the drug. Many of the products of enzymatic breakdown, called metabolites, are less chemically active than the original molecule. For this reason, the liver is referred to as a "detoxifying" organ.
- An example of a Phase I biotransformation is when diazepam, a medication prescribed for anxiety, is transformed into desmethyldiazepam and then to oxazepam. Both these metabolites produce similar physiological and psychological effects of diazepam.
- Catalyzed mainly by **Cytochrome P450 monooxygenases (CYP enzymes)**.

General Reaction:

$$RH + O_2 + NADPH + H^+ \rightarrow ROH + H_2O + NADP^+$$

Example 1: Oxidation of Codeine

$$\text{Codeine} \xrightarrow[\text{CYP2D6}]{\text{O-dealkylation}} \text{Morphine}$$

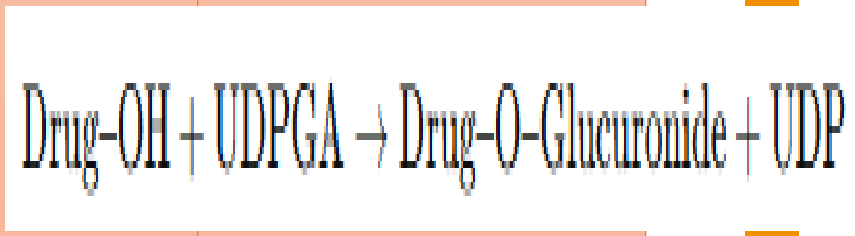
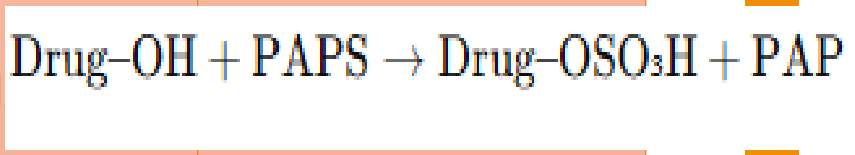
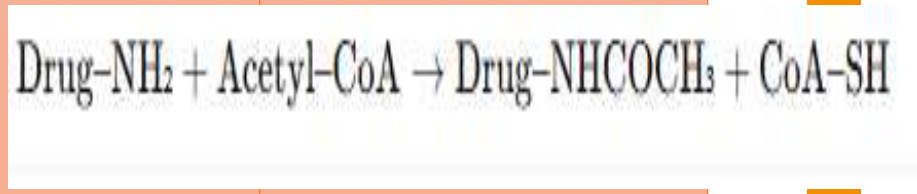
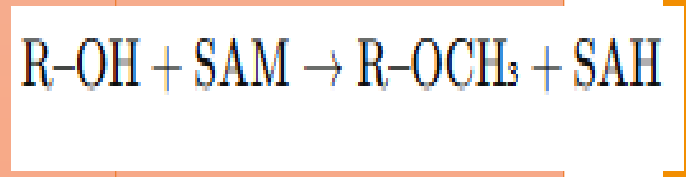
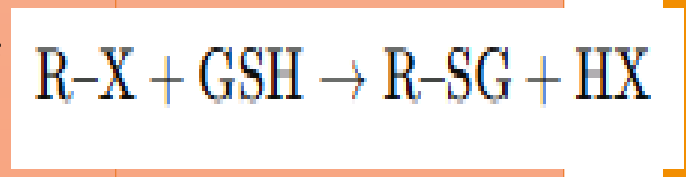
Phase II biotransformations

- Phase II biotransformations involve reactions that couple the drug molecule with another molecule in a process called conjugation.
- Conjugation typically renders the compound pharmacologically inert and water-soluble so it can be easily excreted.
- These processes can occur in the liver, kidney, lungs, intestines, and other organ systems.
- An example of Phase II metabolism is when oxazepam, the active metabolite of diazepam, is conjugated with a molecule called glucuronide so that it becomes physiologically inactive and is excreted without further chemical modification.
- Following Phase II metabolism, Phase III biotransformations may also occur, where the conjugates and metabolites are excreted from cells.

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Type of Conjugation	Co-factor	Example	Reaction
Glucuronidation	UDP-glucuronic acid	Morphine	
Sulfation	PAPS (3'-phosphoadenosine-5'-phosphosulfate)	Paracetamol	
Acetylation	Acetyl-CoA	Isoniazid	
Methylation	S-adenosylmethionine (SAM)	Epinephrine → Metanephrine	
Glutathione conjugation	Glutathione (GSH)	Detoxification of electrophiles	

Cytochrome P450 (CYP) Enzyme System

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- Superfamily of enzymes located in liver microsomes.
- Responsible for oxidative metabolism.
- **CYP3A4, CYP2D6, CYP2C9** are the most significant isoenzymes.
- **Enzyme Inducers:** Rifampicin, Phenobarbital → ↑ metabolism, ↓ efficacy.
  Enzyme Inhibitors: Erythromycin, Ketoconazole → ↓ metabolism, ↑ toxicity.

CYP Reaction Cycle:

$$\text{Drug (RH)} + \text{NADPH} + \text{H}^+ + \text{O}_2 \xrightarrow[\text{CYP}]{} \text{ROH} + \text{H}_2\text{O} + \text{NADP}^+$$

EXCRETION

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- The process by which the body eliminates drugs or their metabolites.
- **Major Routes**
- **Renal excretion** (primary route):
  - **Glomerular filtration:** Free (unbound) drug passes through glomeruli.
  - **Tubular secretion:** Active transport of certain drugs (e.g., penicillin).
  - **Tubular reabsorption:** Lipid-soluble drugs may diffuse back.
- **Biliary excretion:** Large, polar conjugates excreted via bile → intestine → feces.
- **Pulmonary excretion:** For volatile substances (e.g., anesthetic gases).

Renal Excretion Steps

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- **Glomerular Filtration:**
  - Free (unbound) drug filtered through glomeruli.
- Rate = $GFR \times C_{free}$
- **Tubular Secretion:**
  - Active, carrier-mediated transport in proximal tubule.
  - Separate systems for acids (penicillin) and bases (morphine).
- **Tubular Reabsorption:**
  - Lipid-soluble, non-ionized drugs diffuse back into circulation.
  - Ionization depends on **urine pH** (manipulated therapeutically).
- **Ion Trapping Example:**
- To excrete **weak acids** (e.g., salicylates): Alkalinize urine with $NaHCO_3$
  → Drug ionized → Reduced reabsorption → Enhanced excretion.

Pharmacokinetic Parameters

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Parameter	Definition	Importance
Clearance (Cl)	Volume of plasma cleared of drug per unit time	Determines rate of elimination
Half-life (t½)	Time taken for plasma concentration to fall by 50%	Determines dosing interval
Steady-state concentration (Css)	Plasma level where rate of administration = rate of elimination	Achieved after ~4–5 half-lives

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- **Renal Clearance (Cl_n):**

$$Cl_R = \frac{U \times V}{P}$$

where

U = urinary concentration,

V = urine flow rate,

P = plasma concentration.

- **Total Body Clearance (Cl_t):**

$$Cl_T = Cl_{renal} + Cl_{hepatic} + Cl_{other}$$

- **Elimination Half-Life (t½):**

$$t_{1/2} = \frac{0.693 \times V_d}{Cl}$$

The toxicology of commonly used medicinal plants: side effects, cautions and contraindications

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- Toxicology is the study of adverse effects of substances (natural or synthetic) on living organisms.
- Even though medicinal plants are natural, many contain potent bioactive compounds that can cause toxic reactions when used improperly.
- The dose, duration, part of plant used, preparation, and individual sensitivity determine toxicity.

Reasons for Toxicity in Medicinal Plants

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- **Overdosage / prolonged use**
  - → Excess bioactive compounds may cause organ toxicity (e.g., liver, kidney).
- **Adulteration / contamination**
  - → Heavy metals (Pb, Hg, As), pesticides, microbes, or substitution with toxic species.
- **Drug–herb interactions**
  - → Alter metabolism of prescription drugs via CYP enzymes or P-glycoprotein.
- **Intrinsic toxicity**
  - → Some plants naturally contain toxic alkaloids, glycosides, or terpenes.
- **Idiosyncratic reactions**
  - → Hypersensitivity or allergic responses in sensitive individuals.
- **Misidentification of species**
  - → Wrong plant part or similar-looking toxic species used accidentally.

Types of Toxic Effects

Type	Examples	Mechanism
Hepatotoxicity	Pyrrolizidine alkaloids (comfrey), turmeric (rarely)	Hepatocellular necrosis, cholestasis
Nephrotoxicity	Aristolochia species, Aloe latex	Tubular necrosis
Cardiotoxicity	Digitalis, ephedra	Cardiac glycoside toxicity, hypertension
Neurotoxicity	Ginkgo (ginkgotoxin), St. John's Wort	Seizures, serotonin syndrome
Hematologic toxicity	Ginkgo, garlic, ginger	Increased bleeding tendency
Endocrine/reproductive	Ginseng, licorice	Hormonal imbalance, pseudoaldosteronism
Dermatologic/allergic	Aloe (topical), chamomile	Contact dermatitis

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